

NG911 Road Centerline vs PSAP Boundary QA

Location: Pleasant Grove, Alabama

Project Type: NG911 Spatial Data Quality Assurance

Focus: Jurisdictional consistency between road centerlines and PSAP boundary

1. Project Purpose

This project evaluates whether NG911 road centerline features are spatially consistent with the defined PSAP boundary. Road centerlines that fall outside the PSAP boundary can cause call routing ambiguity, jurisdictional confusion, and misdirected emergency response. The goal of this analysis was to identify any road centerline features that do **not** intersect the PSAP boundary.

2. Data Sources

- **RoadCenterlines**
NG911 road centerline dataset provided for Pleasant Grove, AL
 - **PSAP_Boundary**
Official Public Safety Answering Point boundary polygon
 - **Basemap**
Esri World Topographic Map with Hillshade
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3. Methodology

The analysis was performed in **ArcGIS Pro** using a spatial selection workflow:

1. Loaded road centerline and PSAP boundary layers into a project map
2. Verified coordinate systems and layer visibility
3. Used **Select Layer By Location** with the following parameters:
 - **Input Features:** RoadCenterlines

- **Relationship:** Intersect
 - **Selecting Features:** PSAP_Boundary
 - **Invert Spatial Relationship:** Enabled
4. This configuration selects road centerlines that **do not intersect** the PSAP boundary
 5. Reviewed selection results in both the map view and attribute table
 6. Captured a screenshot of the final result for documentation
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4. Results

Selection Result:

- **0 of 807 road centerline features selected**

Interpretation:

No road centerline features were found outside the PSAP boundary.

This indicates full spatial alignment between road centerlines and the PSAP jurisdiction for the study area.

5. QA Finding

Finding: No road centerline features were identified outside the PSAP boundary.

Conclusion: Road centerline jurisdiction alignment appears consistent with PSAP boundaries for Pleasant Grove, Alabama.

This represents a **successful QA validation** with no corrective action required at this stage.

6. Supporting Evidence

- Screenshot documenting zero selected features

File:

RoadCenterlines_Outside_PSAP_NoneFound.jpg

Location:

02_NG911_RoadCenterline_PSAP_QA → 05_Screenshots

7. Significance for NG911 Operations

Although no errors were detected, this analysis confirms:

- Proper jurisdictional coverage of road centerlines
- Reduced risk of call routing errors at PSAP boundaries
- Compliance with NG911 spatial data expectations

Documenting clean datasets is a critical component of operational NG911 quality assurance.

8. Skills Demonstrated

- NG911 spatial data QA methodology
 - ArcGIS Pro spatial selection tools
 - Interpretation of null QA findings
 - Professional documentation of analytical results
 - Folder-based GIS project organization
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9. Notes for Future Work

Future QA checks may include:

- Road centerline parity (left/right address ranges)
- MSAG alignment with road centerlines
- Boundary edge tolerance analysis
- Cross-jurisdictional handoff validation